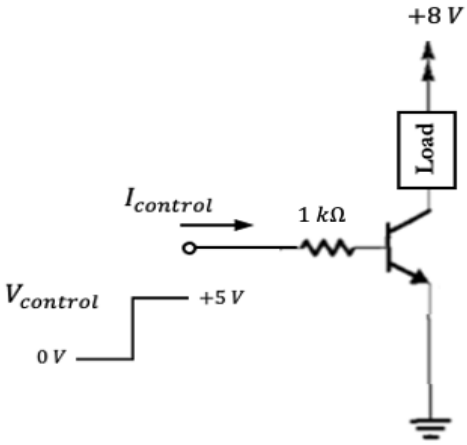
In the given circuit, the BJT has a base-emitter ON voltage of 0.7 V, a base-collector ON voltage of 0.5 V and β=50. Assume that the load is a lamp. The circuit is supposed to switch the lamp ON or OFF based on a digital control signal ($V_{control}$, see figure). Answer the following questions with full details.

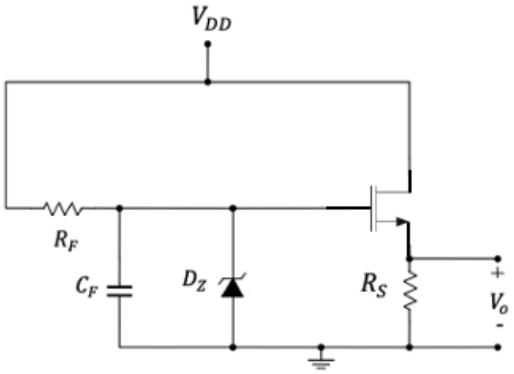


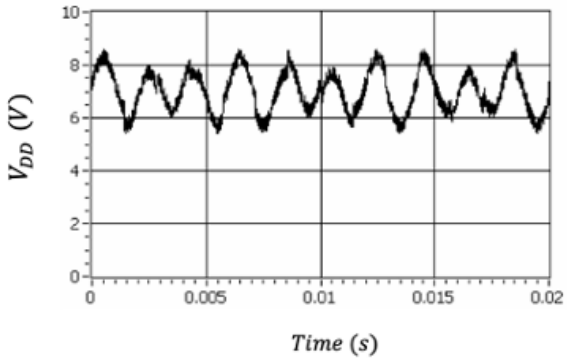
a) The maximum possible voltage <u>across the lamp</u> is (circle one)
(a) +8 V (b) +7.8 V (c) +7.5 V (d) +5 V (e) +4.3 V.
Explain:

b) What are the values of $V_{control}$ and $I_{control}$ to achieve the voltage in part (a)?
c) Show that the minimum lamp resistance to achieve the voltage in part (a) is about 36 Ω.
d) If the lamp resistance is 50 Ω, what is the <u>maximum</u> power consumed by the lamp?
e) If the lamp resistance is 15 Ω, what is the <u>maximum</u> power consumed by the lamp?

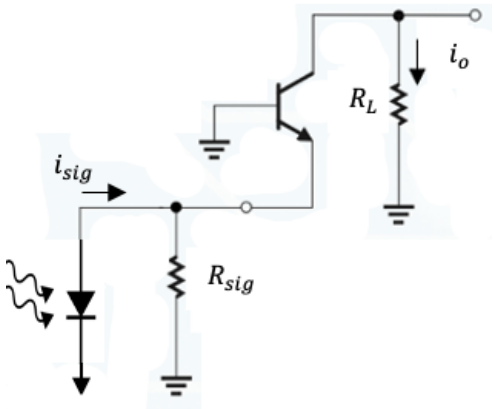
The DC biasing voltage V_{DD} applied to the given circuit has the waveform shown in the figure. The Zener diode has $V_{zo} = 5.3$ V, $r_z = 0\ \Omega$ and $I_{z_min} = 0.5$ mA. The NMOS has $V_t = 1$ V, k=1 mA/V², and λ=0. Take $R_F = 1\ k\Omega$ and $C_F = 1\ \mu F$. Give <u>full details</u> of the solution.

a) On the same graph, sketch the waveform of the output V_o for $R_S = 1\ k\Omega$. Show all labels.
b) Repeat for $R_S = 10\ k\Omega$.





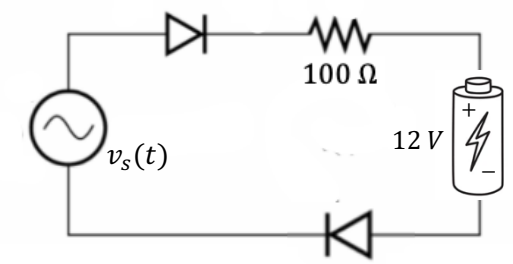
The input to the amplifier circuit below is a weak photodiode current i_{sig} (considered as an independent ac current source). The photodiode has an internal resistance R_{sig}, as shown. Assume that the transistor is in active mode, ignore r_o and,

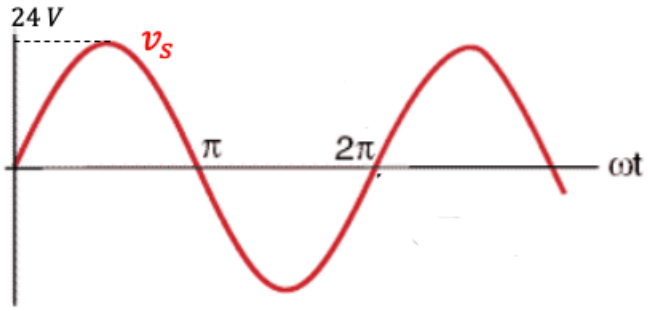


a) Draw the amplifier equivalent circuit using the <u>T-model</u> for the BJT.
b) Show that the amplifier current gain $A_i = \alpha$.
c) Find the overall amplifier current gain.
d) Find the amplifier input resistance.
e) Find the amplifier output resistance.
f) Show that the amplifier voltage gain $A_v = g_m R_L$.
g) Based on your results, does this circuit serve as a "good" current amplifier for this application? Explain.

The ac source v_s applied to the circuit below has the waveform shown in the figure. The diodes have an ON voltage of 0.7 V and $r_D = 0\ \Omega$. Give full details of the solution.

a) On the same graph, sketch the waveform of the current through the rechargeable battery. Show all labels on both axes.
b) Calculate the average power delivered to the rechargeable battery.





Consider the MOSFET amplifier circuit below. Assume that the transistor is in saturation mode, ignore r_o and,

a) Draw the small-signal circuit of the amplifier using the MOSFET pi-model and the T-model.
b) Derive an expression for the voltage gain.
c) Derive an expression for the overall voltage gain.
d) Derive an expression for the current gain.
e) Find the amplifier input resistance.
f) Find the amplifier output resistance.
g) Based on your results, does this circuit serve as a "good" voltage amplifier or as a "good" current amplifier? Explain.

